8

Exergy

Learning Objectives

After reading this chapter you should be able to

- LO 8.1 Define exergy and its utility
- LO 8.2 Summarize the dead state using equilibrium conditions
- LO 8.3 Describe the concept of available exergy and unavailable exergy
- LO 8.4 Interpret why exergy gets degraded when heat flows through a finite temperature difference
- LO 8.5 Evaluate exergy of a gas from a finite energy source and estimate the quality of exergy at any state
- LO 8.6 Describe why second law is called law of degradation of energy
- LO 8.7 Discuss the relation between maximum work done by a closed system in reversible and irreversible process
- LO 8.8 Elaborate how work done in all reversible processes is same working between the same end states
- LO 8.9 Explain the concept of exergy when applied to various systems and differentiate between Helmholtz and Gibbs function
- LO 8.10 Identify the link between irreversibility and Gouy-Stodola theorem and discuss its applications
- LO 8.11 Explain exergy balance and its applications in closed and steady flow systems
- LO 8.12 Define second law efficiency and utilize it in various engineering devices

LO 8.1
Define exergy and its utility

The *exergy* of a system at a given state is the maximum work that can be extracted from it till it reaches the state of thermodynamic equilibrium with its surroundings. The concept of exergy is very important in all energy-producing, energy-consuming and energy-conveying systems. The first law provides an energy analysis of a thermodynamic system without making any discrimination of its quality. The second low emphasises that all forms of energy are not of the same quality and the quality of energy always degrades while conserving its quantity. Both first-law and second-law analyses, i.e. energy and exergy analyses require to be carried out for any energy conversion or energy conveying system, to make it more exergy efficient, which ultimately leads to energy saving or reduced energy consumption for a given task.

8.1 DEAD STATE

LO 8.2
Summarize the dead state using equilibrium conditions

For any substance, if its pressure increases, the volume decreases. Therefore, in any thermodynamic plot in p-v coordinates, all processes must have $\frac{dp}{dv} < 0$, i.e. negative slope. No process having positive slope of dp/dv is possible in nature and cannot be drawn in p-v diagrams. On the other hand, for a substance, if its temperature increases the disorder of its molecules and hence its entropy increase. All processes that have $\frac{dT}{ds} > 0$, i.e. positive slope can only be drawn on T-s diagram. No process having negative slope of $\frac{dT}{ds}$ is possible in nature and cannot be drawn in T-s diagram.

Let p_o and T_o be the ambient pressure and temperature of the surroundings, respectively. If the state of a system departs from that of the surroundings, an opportunity exists for producing work (Fig. 8.1). However, as the system changes its state towards that of the surroundings, this opportunity diminishes, and it ceases to exist when the two are in equilibrium with each other. When the system is in equilibrium with the surroundings, it must be in pressure and temperature equilibrium with the surroundings, i.e. at p_o and T_o. It must also be in chemical equilibrium with the surroundings, i.e. there should not be any chemical reaction or mass transfer. The system must have zero velocity and minimum potential energy. This state of the system is known as the *dead state*, which is designated by affixing subscript 'o' to the properties. It is shown by the point O in Fig. 8.1. Any change in the state of the system from the dead state is the measure of its exergy or maximum work that can be extracted from it. Further, the initial state of the system is from the dead state in terms of p and t, either above or below it, higher will be its exergy value (Fig. 8.1). *All spontaneous processes of a system terminate at the dead state, when* $S = S_{max}$. When a system is at the dead state, there cannot be any spontaneous departure of the system from it, when $S < S_{max}$, and the system must revert to the peak of he entropy hill when $S = S_{max}$ at equilibrium.

Whenever useful work is obtained during a process, in which a finite system undergoes a change of state, the process must terminate when the pressure and temperature of the system have become equal to the pressure and temperature of the surroundings, p_o and T_o, i.e. when the system has reached the dead state. An air engine operating with compressed air taken from a cylinder will continue to deliver work till the pressure of air in the cylinder becomes equal to that of the surroundings, p_o. A certain quantity of exhaust gases from an internal combination engine used as the high temperature source of the heat engine will deliver work till the temperature of the gas becomes equal to that of the surroundings, T_o.

The exergy of a given system is defined as the maximum work that is obtained in a process in which the system comes to equilibrium with its surroundings, i.e. it reaches the dead state. The exergy is thus a composite property depending on the state of both the system and the surroundings. If a system exists at the dead state, its exergy is zero, i.e. it is incapable of delivering any work. Further, it is away from the dead state, either above or below it, higher will be its exergy and more will be its capacity of delivering work.

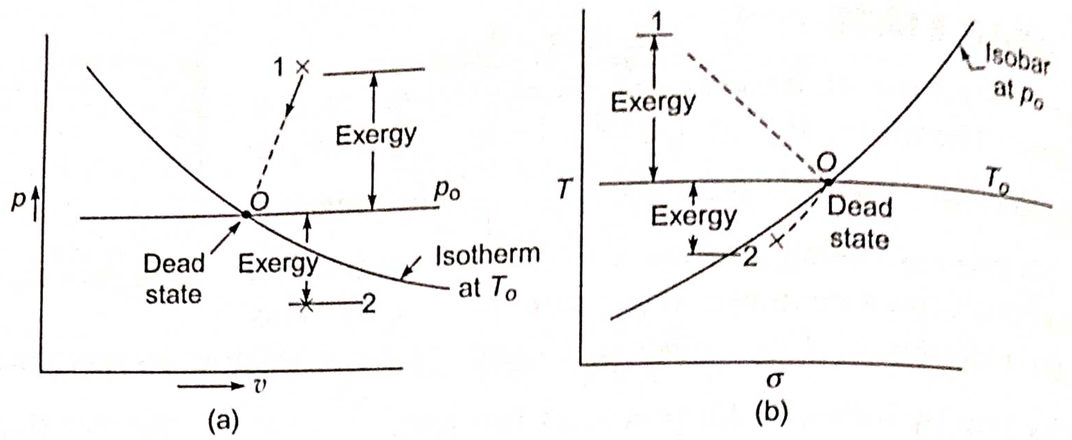

Fig. 8.1 *Exergy of a system decreases as its state approaches the dead state at p_o, T_o, either above or below it*

8.2 EXERGY OF HEAT INPUT IN A CYCLE

LO 8.3 Describe the concept of available exergy and unavailable exergy

The maximum work obtainable from a certain heat input in a cyclic heat engine is often called the available energy (A.E.) or the available part of the energy supplied. The minimum energy that has to be rejected by the second law is called the unavailable exergy (U.E.) or the unavailable part of the exergy supplied.

Let Q be the amount of heat supplied to a reversible engine at temperature T and the heat is rejected to the sink at the dead state T_o (Fig. 8.2). The maximum work obtained or exergy denoted by X is

$$\begin{aligned} X = W_{max} &= \left(1 - \frac{T_o}{T}\right) Q = \text{AE} \\ &= T(s_2 - s_1) - T_o(s_3 - s_4) \\ &= (T - T_o)(s_3 - s_4) \\ &= Q - T_o(s_3 - s_4) \\ &= Q - \text{U.E} = Q - Q_o. \end{aligned}$$

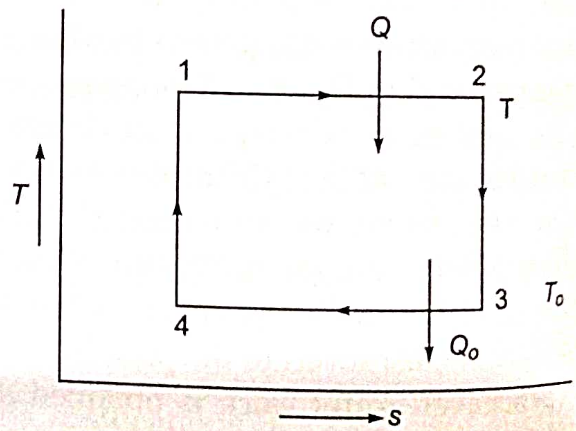

Fig. 8.2 *Exergy of heat input*

The exergy of heat input Q at temperature T is thus equal to $Q \cdot \eta_{rev}$, i.e. $Q\left(1 - \frac{T_o}{T}\right)$. For a given T_o, as T increases, the exergy increases. The exergy decreases as T decreases.

8.3 DECREASE IN EXERGY WHEN HEAT IS TRANSFERRED THROUGH A FINITE TEMPERATURE DIFFERENCE

LO 8.4 Interpret why exergy gets degraded when heat flows through a finite temperature difference

Whenever heat is transferred through a finite temperature difference, there is a decrease in exergy output. Let us consider a reversible heat engine operating between T_1 and T_o (Fig. 8.3). Then

$$Q_1 = T_1\, \Delta s,\ Q_2 = T_o\, \Delta s \text{ and } W = \text{AE} = X = (T_1 - T_o)\, \Delta s$$

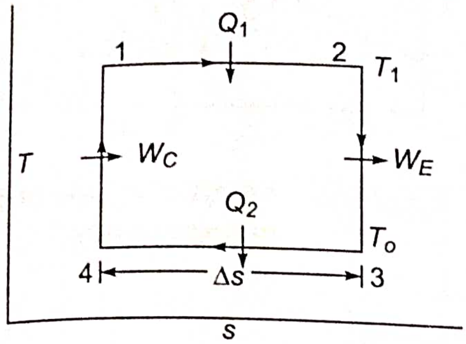

Fig. 8.3 *Carnot cycle*

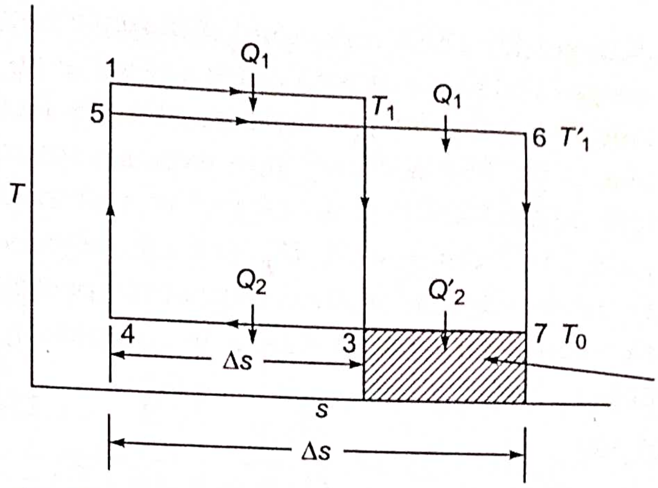

Fig. 8.4 *Decrease in exergy due to heat transfer through a finite temperature difference*

Let us now consider that heat Q_1 is transferred through a finite temperature difference from the reservoir or source at T_1 to the engine absorbing heat at T_1' lower than T_1 (Fig. 8.4). The exergy of Q_1 received by the engine at T_1' can be found by allowing the engine to operate reversibly in a cycle between T_1 and T_o, receiving Q_1 and rejecting Q_2.

Now, $$Q_1 = T_1 \Delta s = T_1' \Delta s'. \text{ Since } T_1 > T_1', \Delta s' > \Delta s$$

$$Q_2 = T_o \Delta s \text{ and } Q_2' = T_o \Delta s'$$

$\therefore$ $$Q_2' > Q_2$$

$$W' = Q_1 - Q_2' = T_1' \Delta s' - T_o \Delta s' = (T_1' - T_o) \Delta s'$$

and $$W = (T_1 - T_o) \Delta s$$

$\therefore$ $$W' < W$$

Exergy lost due to irreversible heat transfer through a finite temperature difference between the source and the working fluid during the heat addition process is given by

$$W - W' = Q_2' - Q_2 = T_o(\Delta s' - \Delta s) \tag{8.1}$$

Exergy destroyed in the process is thus the product of the lowest feasible temperature, i.e. the dead state, and the additional entropy change in the system while receiving heat irreversibly, compared to the case of reversible heat transfer from the same source.

The greater is the temperature difference $(T_1 - T_1')$, the greater is the heat rejection Q_2' and greater will be the exergy lost, or the unavailable part of the energy supplied Q_1. Exergy is said to be destroyed each time it flows through a finite temperature difference.

8.4 EXERGY OF A FINITE BODY AT TEMPERATURE *T*

LO 8.5

Evaluate exergy of a gas from a finite energy source and estimate the quality of exergy at any state

Let us consider a hot gas of mass m_g at temperature T when the environment temperature of the dead state is T_o (Fig. 8.5). Let the gas be cooled at constant pressure from state 1 at temperature T to state 3 at temperature T_o and the heat given up by the gas, Q_1, be utilised in heating up reversibly a working fluid of mass m_{wf} from state 3 to state 1 along the same path so that the temperature difference between the gas and the working fluid at any instant is zero and hence, the entropy increase of the universe is also zero. The working fluid expands reversibly and adiabatically in an engine or turbine from state 1 to 2 doing work W_E, and then rejects heat Q_2 reversibly and isothermally at temperature T_o to return to the initial state 3 to complete a heat engine cycle.

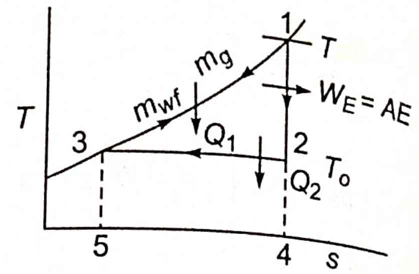

Fig. 8.5 *Exergy of a finite body at temperature T*

Now,

$$Q_1 = m_g c_{p_g} (T - T_o) = m_{wf} c_{p_{wf}} (T - T_o).$$
$$= \text{Area } 14531$$

$\therefore$
$$m_g c_{p_g} = m_{wf} c_{p_{wf}}$$

$\therefore$
$$\Delta S_{gas} = \int_T^{T_o} m_g c_{p_g} \frac{dT}{T} = m_g c_{p_g} \text{ in } \frac{T_o}{T} \text{ (negative)}$$

$$\Delta S_{wf} = \int_T^T m_{wf} c_{p_{wf}} \frac{dT}{T} = m_w c_{p_w} \text{ in } \frac{T}{T_o} \text{ (positive)}$$

$\therefore$
$$\Delta S_{universe} = \Delta S_{gas} + \Delta S_{wf} = 0$$

Also,
$$Q_2 = T_o \Delta S_{wf} = T_o m_{wf} c_{p_{wf}} \ln \frac{T}{T_o}$$

Exergy of the gas is thus

$$X = W_{max} = Q_1 - Q_2$$
$$= m_g c_{p_g} (T - T_o) - T_o m_g c_{p_g} \ln \frac{T}{T_o}$$
$$= m_g c_{p_g} \left[T - T_o - T_o \ln \frac{T}{T_o} \right]$$
$$= \text{Area } 1231$$

Thus the exergy of any finite body of heat capacity C at temperature T is given by

$$X = C\left(T - T_o - T_o \ln \frac{T}{T_o} \right) \tag{8.2}$$

This is similar to Eq. (7.21) as derived from the entropy principle in Chapter 7.

8.5 QUALITY OF ENERGY

Let us assume that a hot gas is flowing through a pipeline (Fig. 8.6). Due to heat loss to the surroundings, the temperature of the gas decreases continuously from inlet at state '*a*' to the exit at state '*b*'. Although the process is irreversible, let us assume

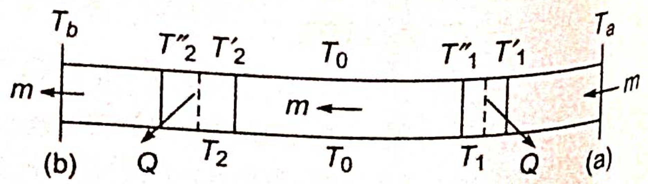

Fig. 8.6 *Heat loss from a hot gas flowing through a pipeline*

a reversible isobaric path between the inlet and exit states of the gas (Fig. 8.7). For an infinitesimal reversible process at constant pressure

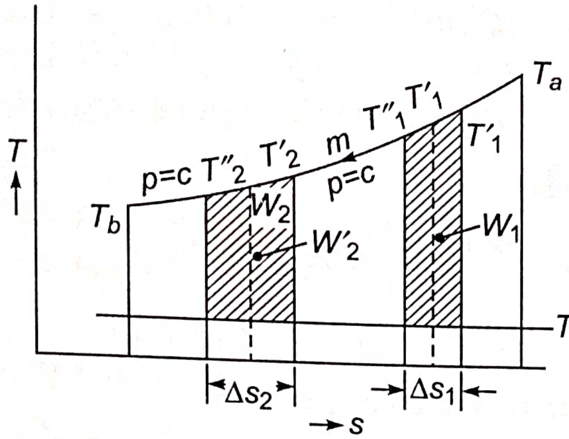

Fig. 8.7 *Degradation of energy quality of gas as temperature of the gas decreases*

$$dS = \frac{mc_p dt}{T}$$

$$\therefore \qquad \frac{dT}{dS} = \frac{T}{mc_p} \tag{8.3}$$

where m is the mass of gas flowing and c_p is its specific heat. The slope $\frac{dT}{dS}$ depends on the gas temperature T. As T increases, the slope increases and if T decreases, slope decreases.

Let us assume that Q units of heat are lost to the surroundings at T_o as the temperature of the gas decreases from T_1' to T_1'', T_1 being the average of the two. Then

Heat loss, $$Q = mc_p (T_1' - T_1'') = T_1 \Delta S_1 \tag{8.4}$$

Exergy of this heat loss at temperature T_1 is

$$W_1 = Q - T_o \Delta S_1 \tag{8.5}$$

When the gas temperature has reached T_2 ($T_2 < T_1$), let us assume that the same heat loss Q occurs as the gas temperature decreases from T_2' to T_2'', T_2 being the average temperature of the gas.

Heat loss, $$Q = mc_p (T_2' - T_2'') = T_2 \Delta S_2 \tag{8.6}$$

Exergy of this heat loss at temperature T_2 is

$$W_2 = Q - T_o \Delta S_2 \tag{8.7}$$

From Eqs. (8.4) and (8.6),

Since $T_1 > T_2$, $\Delta S_1 < \Delta S_2$, therefore, from Eqs. (8.5) and (8.7),

$$W_1 > W_2 \tag{8.8}$$

The exergy of the gas at T_1 is more than that at T_2.

The loss of exergy is more when heat loss occurs at a higher temperature T_1 than when the same heat loss occurs at a lower temperature T_2. Therefore, a heat loss of 1 kJ at, say, 1000°C is more harmful than the same heat loss of 1 kJ at, say, 100°C. The exergy of a system at temperature T is given by Eq. (8.2),

$$X = C\left(T - T_o - T_o \ln \frac{T}{T_o}\right)$$

At $T = 1000 + 273 = 1273$ K and $T_o = 300$ K,

$$X_1 = C\left(1273 - 300 - 300 \ln \frac{1273}{300}\right)$$

$$= C(973 - 433.6) = 539.4\ C \text{ kJ}$$

At $T = 100 + 273 = 373$ K,

$$X_2 = C\left(373 - 300 - 300 \ln \frac{373}{300}\right)$$

$$= C(73 - 65.34) = 7.66\ C \text{ kJ}$$

where C is the heat capacity of the body in kJ/K.

Therefore, $X_1 > X_2$. Adequate insulation must be provided for high temperature fluids ($T >> T_o$) to reduce the precious heat loss. This may not be so important for low temperature fluids ($T \sim T_o$), since the loss of exergy from such fluids would be low. The cost of insulation may be more than the money saved by reducing the heat loss. Similarly, insulation must be provided adequately for very low temperature fluids ($T << T_o$) whose exergy is high to reduce heat gain from the surroundings and preserve its exergy.

We have seen that the exergy of a fluid at a higher temperature T_1 is more than that at a lower temperature T_2, and it decreases as the temperature decreases.

The second law thus affixes a quality to energy of a system at any state. The quality of energy of a gas at, say, 1000°C is superior to that at, say, 100°C. An awareness of this energy quality as of energy quantity is essential for the efficient use of our energy resources and for energy conservation. *The concept of exergy provides a useful measure of this energy quality.*

8.6 LAW OF DEGRADATION OF ENERGY

LO 8.6
Describe why second law is called law of degradation of energy

The exergy of a system decreases as its temperature or pressure decreases and approaches that of the dead state where its value is zero. When heat is transferred from a system, its temperature decreases and the quality of its energy deteriorates and its exergy decreases. The degradation of energy of the system is more when heat loss occurs at a higher temperature than that occurring at a lower temperature. Quality wise the energy loss may be the same, but quantity wise the losses are different. *While the first law states that energy is always conserved quantity wise, the second law emphasises that energy is always degraded quality wise.* When a gas is throttled adiabatically from high to low pressure, the enthalpy or energy of gas per unit mass remains the same, but there is a degradation of it energy or capacity of doing work. The same holds good for pressure drop due to friction of a fluid flowing through an insulated pipe. If the first law is the law of conservation of energy, the second law is called *the law of degradation of energy.*

Energy is always conserved, but its quality is always degraded.

8.7 MAXIMUM WORK IN A REVERSIBLE PROCESS

LO 8.7
Discuss the relation between maximum work done by a closed system in reversible and irreversible process

Let us consider a closed stationary system undergoing a reversible process R from state 1 to 2 interacting with the surroundings at p_o, To (Fig. 8.8). Then by the first law,

$$Q_R = U_2 - U_1 + W_R \tag{8.9}$$

If the process were irreversible, as represented by the dotted line I connecting the same equilibrium points 1 and 2,

$$Q_I = U_2 - U_1 + W_I \tag{8.10}$$

Therefore, from Eqs. (8.9) and (8.10),

$$Q_R - Q_I = W_R - W_I \tag{8.11}$$

Now,

$$\Delta S_{\text{sys}} = S_2 - S_1$$

and

$$\Delta S_{\text{surr}} = -\frac{Q}{T_o}$$

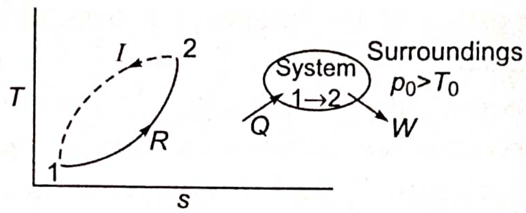

Fig. 8.8 *Maximum work done by a closed system*

By second law,

$$\Delta S_{\text{univ}} \geqslant O$$

For a reversible process,

$$\Delta S_{\text{univ}} = S_2 - S_1 - \frac{Q_R}{T_o} = O$$

$$Q_R = T_o(S_2 - S_1) \tag{8.12}$$

For an irreversible process,

$$\Delta S_{\text{univ}} > O$$

or

$$S_2 - S_1 - \frac{Q_I}{T_o} > O$$

$\therefore$

$$Q_I < T_o(S_2 - S_1) \tag{8.13}$$

From Eqs. (8.12) and (8.13),

$$Q_R > Q_1$$

$\therefore$ From Eq. (8.11),

$$W_R > W_I \tag{8.14}$$

The work done by a closed system by interacting only with the surroundings at p_o, T_o in a reversible process is always more than that done by it in an irreversible process between the same end states.

8.8 WORK DONE IN ALL REVERSIBLE PROCESSES IS THE SAME

LO 8.8
Elaborate how work done in all reversible processes is same working between the same end states

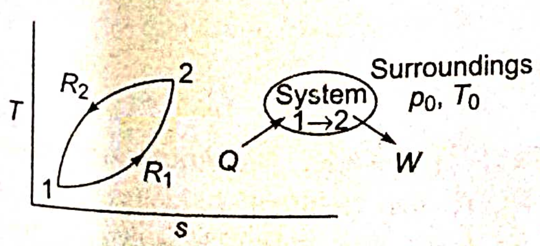

Fig. 8.9 *Equal work done in all reversible process between the same end states*

Let us assume two reversible processes R_1 and R_2 between the same end states 1 and 2 undergone by a closed system by exchanging energy only with the surroundings (Fig. 8.9). Let one of the processes be reversed.

Then the system would execute a reversible cycle 1-2-1 and produces net work represented by the area enclosed by exchanging energy with only one reservoir, i.e. the surroundings. This violates the Kelvin–Planck statement. Therefore, *the two reversible processes must coincide and produce equal amounts of work.*

8.9 EXERGY OF A CLOSED SYSTEM

LO 8.9 Explain the concept of exergy when applied to various systems and differentiate between Helmholtz and Gibbs function

For a closed system having energy interactions Q and W and undergoing change of state from 1 to 2 (Fig. 8.10) and $T_o > T$ so that heat flows into the system, from the surroundings at T_o

$$Q = E_2 - E_1 + W$$

$$\therefore \quad W = E_1 - E_2 + Q \tag{8.15}$$

where E is the energy of the system. The temperature of the portion of the boundary across which heat transfer is taking place is denoted by T. Since $T_o > T$, the heat transfer is obviously irreversible. To make the heat transfer process reversible, let a Carnot engine be operated between T_o and T receiving heat Q_o from the surroundings, producing work W_c and rejecting heat Q at T, which enters the system. The maximum work delivered by the system will be

$$W_{max} = W + W_C - p_o(V_2 - V_1) \tag{8.16}$$

where $p_o(V_2 - V_1)$ is the work that will be done by the system against the surroundings having a flexible boundary with the volume of the system increasing from V_1 to V_2 (which will be zero if the boundary is rigid). The work done by the Carnot engine,

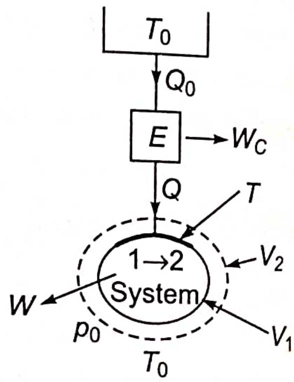

Fig. 8.10 *Maximum work done by a closed system with heat flowing into the system ($T_o > T$)*

$$W_C = Q_o - Q$$

For a reversible engine, $\dfrac{Q_o}{T_o} = \dfrac{Q}{T} = S_2 - S_1$

$$\therefore \quad W_C = T_o\frac{Q}{T} - Q, \tag{8.17}$$

From Eqs. (8.15), (8.16) and (8.17), neglecting K.E. and P.E. terms (when $E = U$),

$$W_{max} = U_1 - U_2 + Q + T_o\frac{Q}{T} - Q - p_o(V_2 - V_1)$$

$$= U_1 - U_2 + T_o(S_2 - S_1) - p_o(V_2 - V_1)$$

$$= U_1 - U_2 + p_o(V_1 - V_2) - T_o(S_1 - S_2) = (W_u)_{max} \tag{8.18}$$

This is the maximum useful work $(W_u)_{max}$ delivered by the system.

If heat is rejected from the system across the boundary where the temperature is T and $T > T_o$, the first law gives (Fig. 8.11), neglecting K.E. and P.E. terms,

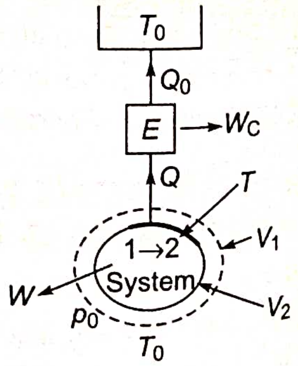

Fig. 8.11 *Maximum work done by a closed system with heat rejection from the system ($T > T_o$)*

$$-Q_1 = U_2 - U_1 + W$$

$$\therefore \quad W = U_1 - U_2 - Q \tag{8.19}$$

The work done by the reversible engine operating between T and T_o.

$$W_C = Q - Q_o$$

By the second law, $\frac{Q_o}{T_o} = \frac{Q}{T}$ and $-\frac{Q}{T} = S_2 - S_1$ (8.20)

$$\therefore \quad \dot{W}_{max} = W + W_C - p_o(V_2 - V_1)$$

$$= U_1 - U_2 - Q + Q - Q_o - p_o(V_2 - V_1)$$

$$= U_1 - U_2 + T_o\frac{Q}{T} - p_o(V_1 - V_2) = U_1 - U_2 + T_o(S_2 - S_3) - p_o(V_2 - V_1)$$

$$= U_1 - U_2 + p_o(V_1 - V_2) - T_o(S_1 - S_2)$$

This is the same as given by Eq. (8.18), which provides the maximum work done by the closed system, also called *availability* A, in the change of state from 1 to 2 by interacting only with the surroundings at p_o, T_o. This expression can also be written in the alternate from

$$W_{max} = A = (U_1 + p_oV_1 - T_oS_1) - (U_2 + p_oV_2 - T_oS_2) \tag{8.21a}$$

$$= \phi_1 - \phi_2 \tag{8.21b}$$

where ϕ_1 and ϕ_2 are exergy functions of a closed system at states 1 and 2, respectively. For unit mass of substance

$$W_{max} = a = (u_1 + p_ov_1 - T_os_1) - (u_2 + p_ov_2 - T_os_1)$$

$$= (u_1 - u_2) + p_o(v_1 - v_2) - T_o(s_1 - s_2)$$

With reference to the dead state, the maximum work that can be delivered by the closed system at p, T is given by

$$W_{max} = A = (U - U_o) + p_o(V - V_o) - T_o(S - S_o)$$

$$= (U + p_oV - T_oS) - (U_o + p_oV_o - T_oS_o)$$

$$= \phi - \phi_o \tag{8.22}$$

where $\phi_o = U_o + p_oV_o - T_oS_o$ is the exergy function of the dead state. For unit mass of a substance,

$$W_{max} = a = (u + p_ov - T_os) - (u_o + p_ov_o - T_os_o)$$

$$= \phi - \phi_o$$

8.10 EXERGY OF A STEADY FLOW SYSTEM

For an open or a steady flow system (Fig. 8.12), neglecting kinetic and potential energy changes, the steady flow energy equation (S.F.E.E.) gives

$$H_1 + Q = H_2 + W$$

or

$$W = H_1 - H_2 + Q \tag{8.23}$$

Now, with a reversible heat engine E operating between T_o and T, the work done will be

$$W_C = Q_o - Q$$

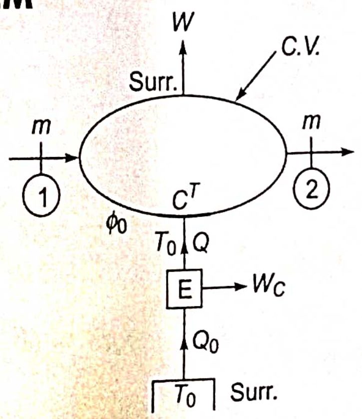

Fig. 8.12 *Maximum work done by a steady flow system*

Also, $\dfrac{Q}{T} = \dfrac{Q_o}{T_o}$ and $\dfrac{Q}{T} = S_2 - S_1$ (8.24)

∴ Maximum work obtained from the steady flow system would be from Eqs. (8.23) and (8.24), also called availability, A_f

$$W_{Max} = A_f = W + W_c$$
$$= H_1 - H_2 + Q + Q_o - Q$$
$$= H_1 - H_2 + Q.\frac{T_o}{T} = H_1 - H_2 + T_o(S_2 - S_1)$$
$$= H_1 - H_2 - T_o(S_1 - S_2)$$
$$= (H_1 - T_oS_1) - (H_2 - T_oS_2) = B_1 - B_2 \tag{8.25}$$

where $B = H - T_oS$ is called the *Keenan function.*

For unit mass of a substance,

$$W_{max} = a_f = (h_1 - h_2) - T_o(s_1 - s_2)$$
$$= (h_1 - T_os_1) - (h_2 - T_os_2)$$
$$= b_1 - b_2 \quad (8.26)$$

If K.E. and P.E. terms are taken into consideration, Eqs. (8.25) and (8.26) become

$$W_{max} = A_f = \left(H_1 + \frac{mV_1^2}{2} + mgZ_1 - T_oS_1\right) - \left(H_2 + \frac{mV_2^2}{2} + mgZ_2 - T_oS_2\right)$$
$$= \left(B_1 + \frac{mV_1^2}{2} + mgZ_1\right) - \left(B_2 + \frac{mV_2^2}{2} + mgZ_2\right)$$
$$= \psi_1 - \psi_2 \tag{8.27}$$

where $\psi = B + \dfrac{mV^2}{2} + mgZ$ is the *exergy function of a steady flow system.*

With reference to the dead state (Fig. 8.13), Eqs. (8.26) and (8.27) become

$$W_{max} = a_f = (h - T_os) - (h_o - T_os_o)$$
$$= (h - h_o) - T_o(s - s_o)$$
$$= b - b_o \tag{8.28}$$

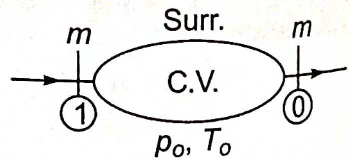

Fig. 8.13 *Exergy of a steady flow system*

$$W_{max} = a_f = \left(h + \frac{V^2}{2} + gZ\right) - (h_o + gZ_o) - T_o(s - s_o)$$
$$= (h - h_o) + \frac{V^2}{2} + g(Z - Z_o) - T_o(s - s_o)$$
$$= \left(h - T_oS + \frac{V^2}{2} + gZ\right) - (h_o - T_os_o + qZ_o)$$
$$= \left(B + \frac{V^2}{2} + gZ\right) - (b_o + gZ_o)$$
$$= \psi - \psi_o \tag{8.29}$$

since the velocity of the system at the dead state is zero.

8.11 OBTAIN MAXIMUM WORK WHEN THE SYSTEM EXCHANGES HEAT WITH A THERMAL RESERVOIR IN ADDITION TO THE ATMOSPHERE

If the open system as discussed in Section 8.9 exchanges heat with a thermal reservoir at temperature T_R in addition to the atmosphere, the maximum work will be increased by $Q_R\left(1-\frac{T_o}{T_R}\right)$,

i.e. $$W_{\max} = \left(H_1 - T_oS_1 + \frac{mV_1^2}{2} + mgZ_1\right) - \left(H_2 - T_oS_2 + \frac{mV_2^2}{2} + mgZ_2\right) + Q_R\left(1-\frac{T_o}{T_R}\right)$$

$$= \psi_1 - \psi_2 + Q_R\left(1-\frac{T_o}{T_R}\right) \tag{8.30}$$

For a closed system,

$$W_{\max} = E_1 - E_2 + p_o(V_1 - V_2) - T_o(S_1 - S_2) + Q_R\left(1-\frac{T_o}{T_R}\right) \tag{8.31}$$

If K.E. and P.E. terms are neglected, for a steady flow system

$$W_{\max} = H_1 - H_2 - T_o(S_1 - S_2) + Q_R\left(1-\frac{T_o}{T_R}\right)$$

and for a closed non-flow system,

$$W_{\max} = U_1 - U_2 + p_o(V_1 - V_2) - T_o(S_1 - S_2) + Q_R\left(1-\frac{T_o}{T_R}\right) \tag{8.32}$$

8.12 EXERGY IN CHEMICAL REACTIONS

In many chemical reactions, the reactants are often in pressure and temperature equilibrium with the surroundings (before the reaction takes place) and so are the products after the reaction. An internal combustion engine can be cited as an example of such a process if we visualise the products being cooled to atmospheric temperature T_o before being discharged from the engine.

1. Let us consider a system which is in temperature equilibrium with the surroundings before and after the process. The maximum work obtainable during a change of state is given by Eq. (8.18),

$$W_{\max} = E_1 - E_2 - T_o(S_1 - S_2)$$

$$= \left(U_1 + \frac{mV_1^2}{2} + mgZ_1\right) - \left(U_2 + \frac{mV_2^2}{2} + mgZ_2\right) - T_o(S_1 - S_2)$$

If K.E. and P.E. terms are neglected

$$W_{\max} = U_1 - U_2 - T_o(S_1 - S_2)$$

Since the initial final temperature are the same as that of the surroundings,

$$T_1 = T_2 = T_o = T, \text{ say, then}$$

$$W_{\max} = (U_1 - U_2)_T - T(S_1 - S_2)_T \tag{8.33}$$

Let a property called *Helmholtz functions* F be defined by the relation

$$F = U - T_5 \tag{8.34}$$

Thus for two equilibrium states 1 and 2 at the same temperature T,

$$(F_1 - F_2)_T = (U_1 - U_2)_T - T(S_1 - S_2)_T \tag{8.35}$$

From Eqs. (8.34) and (8.35)

$$(W_T)_{max} = (F_1 - F_2)_T \tag{8.36}$$

Therefore,

$$W_T \leqslant (F_1 - F_2)_T \tag{8.37}$$

The work done by a system in any process between two equilibrium states at the same temperature during which the system exchanges heat only when the environment is equal to or less than the decrease in the Helmholtz function of the system during the process. The maximum work is done when the process is reversible and the equality sign holds. If the process is irreversible, the work is less than the maximum.

2. Let us now consider a system which is in both pressure and temperature equilibrium with the surroundings before and after the process. When the volume of the system increases, same work is done by the system against the surroundings (pdV work) and this is not available for doing useful work. The exergy of the system, as defined by Eq. (8.22), neglecting the K.E. and P.E. terms, can be expressed in the form,

$$\begin{aligned} A &= (W_u)_{max} = (U + p_o V - T_o S) - (U_o + p_o V - T_o S_o) \\ &= \phi - \phi_o \end{aligned}$$

The maximum work obtainable during a change of state is the decrease of its exergy given by

$$\begin{aligned} (W_u)_{max} &= A_1 - A_2 = \phi_1 - \phi_2 \\ &= (U_1 - U_2) + p_o (V_1 - V_2) - T_o(S_1 - S_2) \end{aligned}$$

When the initial and final equilibrium states of the system are at the same pressure and temperature of the surroundings, say,

$$p_1 = p_2 = p_o = p \quad \text{and} \quad T_1 = T_2 = T_o = T, \text{ then}$$

$$(W_u)_{max} = (U_1 - U_2)_{P,T} + p(V_1 - V_2)_{P,T} - T(S_1 - S_2)_{P,T} \tag{8.38}$$

The *Gibbs function* F is defined as

$$\begin{aligned} G &= H - TS \\ &= U + pV - TS \end{aligned} \tag{8.39}$$

Thus for two equilibrium states at the same pressure p and temperature T

$$(G_1 - G_2)_{P,T} = (U_1 - U_2)_{P,T} + P(V_1 - V_2)_{P,T} - T(S_1 - S_2)_{P,T} \tag{8.40}$$

From Eqs. (8.38) and (8.40),

$$(W_u)_{max} = (G_1 - G_2)_{P,T}$$

$$\therefore \quad (W_u)_{P,T} \leqslant (G_1 - G_2)_{P,T} \tag{8.41}$$

The decrease in the Gibbs function of a system sets an upper limit to the work that can be performed, exclusive of the pdV work, in any process between two equilibrium points at the same pressure and temperature, provided the system exchanges heat only with the environment which is at the same temperature and pressure as the end states of the system. If the process is irreversible, the useful work is less than the maximum.

8.13 IRREVERSIBILITY AND GOUY–STODOLA THEOREM

LO 8.10
Identify the link between irreversibility and Gouy-Stodola theorem and discuss its applications

The actual work done by a system is always less than the idealised reversible work, and the difference between the two is called the *irreversibility* (I) of the process.

$$\therefore \quad I = W_{rev} - W = W_{max} - W \qquad (8.42)$$

This is also often referred to as '*degradation*' or '*dissipation*'.

For a non-flow process between two equilibrium states, when the system exchanges heat only with the surroundings (Fig. 8.14), neglecting K.E. and P.E. terms, the first law gives from Eq. (8.42),

$$\begin{aligned} I &= [(U_1 - U_2) - T_o(S_1 - S_2)] - [(U_1 - U_2) + Q] \\ &= T_o(S_2 - S_1) - Q \\ &= T_o(\Delta S)_{system} + T_o(\Delta S)_{surr.} \\ &= T_o(\Delta S)_{univ} \qquad (8.43) \\ \therefore \quad I &\geqslant O \end{aligned}$$

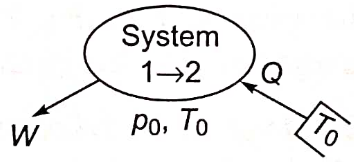

Fig. 8.14 *Irreversibility in a process*

Similarly, for the steady flow process

$$\begin{aligned} I &= W_{max} - W \\ &= \left(B_1 + \frac{mV_1^2}{2} + mgZ_1\right) - \left(B_2 + \frac{mV_2^2}{2} + mgZ_2\right) \\ &- \left(H_1 + \frac{mV_1^2}{2} + mgZ_1\right) - \left(H_2 + \frac{mV_2^2}{2} + mgZ_2\right) \\ &= T_o(S_2 - S_1) - Q \\ &= T_o(\Delta S)_{sys} + T_o(\Delta S)_{surr} = T_o(\Delta S)_{univ.} \qquad (8.44) \end{aligned}$$

The same expression for irreversibility holds good for both flow and non-flow processes. The quantity $T_o(\Delta S)_{univ}$ or $T_o S_{gen}$ represents the decrease in exergy of the system due to irreversibility.

This relation, Eq. (8.44), is called the *Gouy–Stodola theorem* which states that the rate of loss of exergy in a process is proportional to the rate of *entropy generation*. If (Eq. (8.44)) is written in the rate form

$$\dot{I} = \dot{W}_{lost} = T_o(\Delta \dot{S})_{univ} = T_o\dot{S}_{gen} \qquad (8.45)$$

This is known as *Gouy–Stodola equation*. A thermodynamically efficient process would involve minimum exergy loss with minimum rate of entropy generation.

8.13.1 Application of Gouy–Stodola Equation

1. Heat transfer through a finite temperature difference If heat transfer $\dot{Q}$ occurs from the hot reservoir at T_1 to the cold reservoir at T_2 (Fig. 8.15), the rate of entropy generation

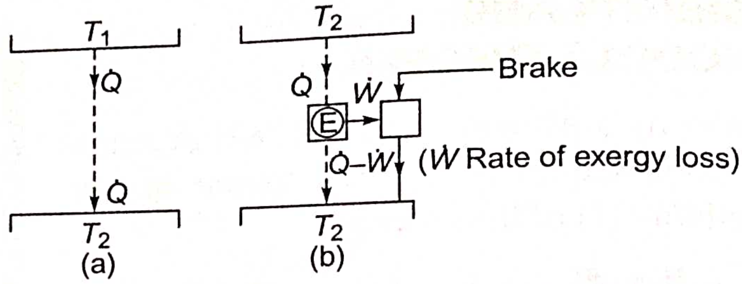

Fig. 8.15 *Decrease of exergy by heat transfer through a finite temperature difference*

$$\dot{S}_{gen} = \frac{\dot{Q}}{T_2} - \frac{\dot{Q}}{T_1} = \dot{Q}\frac{T_1 - T_2}{T_1 T_2}$$

and
$$\dot{W}_{lost} = T_2 \dot{S}_{gen} = \dot{Q}\frac{T_1 - T_2}{T_1} = \dot{Q}.\eta_{rev} \tag{8.46}$$

If the heat transfer $\dot{Q}$ from T_1 to T_2 takes place through a reversible engine, work output $\dot{W}$ is dissipated in the brake, from which an equal amount of heat is rejected to the reservoir at T_2 (Fig. 8.15 b). Heat transfer through a finite temperature difference is equivalent to the destruction of its exergy.

2. Flow with friction Let us consider the steady and adiabatic flow of an *ideal gas* through the segment of a pipe (Fig. 8.16a), by the first law

$$h_1 = h_2$$

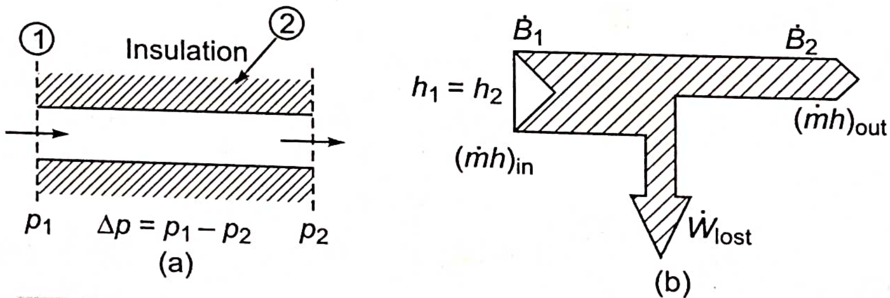

Fig. 8.16 *Irreversibility in a duct flow due to fluid friction*

and by the second law

$$Tds = dh - vdp$$

Since for an ideal gas, enthalpy is a function of temperature, which here remains constant, $dh = 0$, the entropy change

$$\int_1^2 ds = -\int_1^2 \frac{vdp}{T} = -\int_1^2 R\frac{dp}{p}$$

and the rate of entropy generation

$$\dot{S}_{gen} = \int_1^2 \dot{m}\, ds = -\int_1^2 \dot{m} R \frac{dp}{p} = \dot{m} R \ln \frac{p_1}{p_2}$$

$$= -\dot{m} R \ln\left(1 - \frac{\Delta p}{p_1}\right) = -\dot{m}R\left(-\frac{\Delta p}{p_1}\right)$$

$$= \dot{m}R\frac{\Delta p}{p_1} \tag{8.47}$$

where $\ln\left(1 - \frac{\Delta p}{p_1}\right) = -\frac{\Delta p}{p_1}$, since $\frac{\Delta p}{p_1} < 1$.

(We know: ln $(1-x) = -x - \frac{x^3}{3} - \frac{x^5}{5} - \dots = -x$, where $x < 1$, with higher terms being neglected)

$$\dot{W}_{\text{lost}} = \dot{B}_1 - \dot{B}_2$$
$$= \dot{m}\,[(h_1 - T_o s_1) - (h_2 - T_o s_2)]$$
$$= \dot{m} T_o (s_2 - s_1)$$
$$= T_o \dot{S}_{\text{gen}} = \dot{m}\, R T_o \frac{\Delta p}{p} \tag{8.48}$$

The rate of exergy loss, $\dot{W}_{\text{lost}}$, is thus proportional to the pressure drop (Δp) and the mass flow rate ($\dot{m}$). It is shown on the right (Fig. 8.16b) by the *Grassmann diagram*, the width at any section being proportional to the *exergy* of the fluid (system). It is an adaptation of the *Sankey diagram* where the width at any section is proportional to the *energy* of the fluid in a pipeline.

3. Mixing of two fluids Two streams 1 and 2 of an incompressible fluid or an ideal gas mix adiabatically at constant pressure (Fig. 8.17).

Here, $\dot{m}_1 + \dot{m}_2 = \dot{m}_3 = \dot{m}$ (say)

Let $x = \dfrac{\dot{m}}{\dot{m}_1 + \dot{m}_2}$

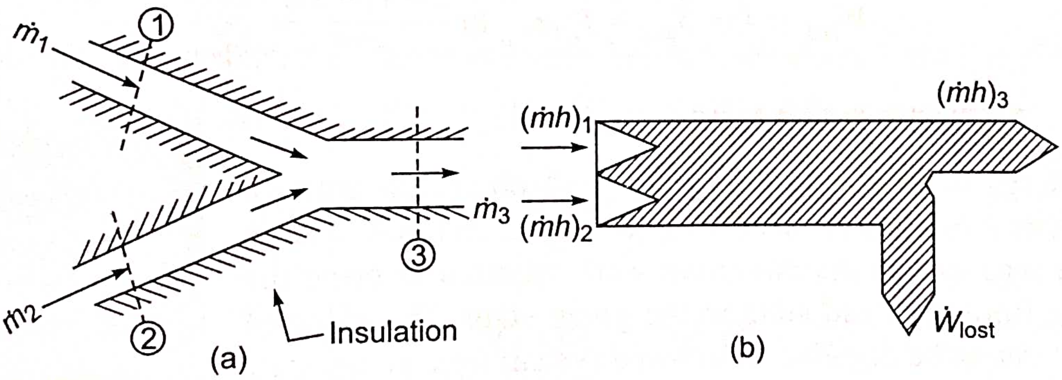

Fig. 8.17 *Rate of exergy loss due to mixing*

By the first law,

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_2)\, h_3$$
$$x h_1 + (1-x) h_2 = h_3$$

The preceding equation can be written in the following form since enthalpy is a function of temperature

$$x T_1 + (1-x) T_2 = T_3$$

or
$$\frac{T_3}{T_1} = x + (1-x)\tau \tag{8.49}$$

where $\tau = T_2/T_1$

By the second law,

$$\dot{S}_{\text{gen}} = \dot{m}_3 s_3 - \dot{m}_1 s_1 - \dot{m}_2 s_2$$

or
$$\dot{S}_{\text{gen}} = \dot{m} s_3 - x \dot{m} s_1 - (1-x)\, \dot{m} s_2$$

$$\therefore \quad \frac{\dot{S}_{\text{gen}}}{\dot{m}} = (s_3 - s_2) + x(s_2 - s_1).$$

$$= c_p \ln \frac{T_3}{T_2} + x c_p \ln \frac{T_2}{T_1}$$

$$\frac{\dot{S}_{\text{gen}}}{\dot{m}c_p} = \ln \frac{T_3}{T_2} + x \ln \frac{T_2}{T_1}$$

$$= \ln \left(\frac{T_3}{T_2}\right)\left(\frac{T_2}{T_1}\right)^x$$

or

$$N_s = \ln \frac{T_3}{T_1}\frac{T_1}{T_2}\left(\frac{T_2}{T_1}\right)^x$$

$$= \ln \frac{T_3/T_1}{(T_2/T_1)^{1-x}} = \frac{x + (1-x)z}{z^{1-x}} \tag{8.50}$$

where N_S is a dimensionless quantity, called the *entropy generation number*, given by $\frac{\dot{S}_{\text{gen}}}{\dot{m}c_p}$. If $x = 1$ or $\tau = 1$, N_s becomes zero in each case. The rate of exergy loss due to mixing would be

$$\dot{W}_{\text{lost}} = \dot{I} = \dot{S}_{\text{gen}} = T_o\, \dot{m}\, c_p \ln \frac{x + (1-x)\tau}{\tau^{1-x}} \tag{8.51}$$

8.14 EXERGY BALANCE

LO 8.11 Explain exergy balance and its applications in closed and steady flow systems

The exergy is the maximum useful work obtainable from a system as it reaches the dead state (p_o, T_o). Conversely, exergy can be regarded as the minimum work required to bring the system from the dead state to the given state. The value of exergy cannot be negative. If a closed system were at any state other than the dead state, the system would be able to change its state spontaneously towards the dead state. This tendency would stop when the dead state is reached. No work is done to effect such a spontaneous change. Since any change in state of the closed system to the dead state can be accomplished with zero work, the *maximum work or exergy cannot be negative*.

While energy is always conserved, exergy is not generally conserved, but it is destroyed by irreversibilities. When a closed system is allowed to undergo a spontaneous change from the given state to the dead state, its exergy is completely destroyed without producing any useful work. The potential to develop work that exists originally at the given state is thus completely wasted in such a spontaneous process.

Therefore, at steady state

1. Energy in – Energy out = 0
2. Exergy in – Exergy out = Exergy destroyed

8.14.1 Exergy Balance for a Closed System

For a closed system, exergy transfer occurs through heat and work interactions (Fig. 8.18).

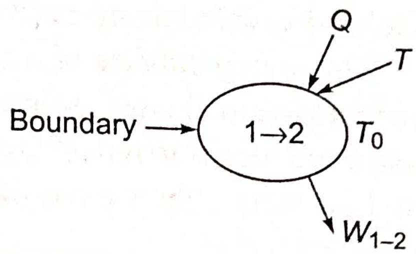

Fig. 8.18 *Exergy balance for a closed system*

First law: $$E_2 - E_1 = \int_1^2 đQ - W_{1-2} \qquad (8.52)$$

Second law: $$S_2 - S_1 - \int_1^2 \frac{đQ}{T} = S_{\text{gen}}$$

or $$T_o(S_2 - S_1) - T_o \int_1^2 \frac{đQ}{T} = T_o S_{\text{gen}} \qquad (8.53)$$

Subtracting Eq. (8.53) from (Eq. 8.52),

$$E_2 - E_1 - T_o(S_2 - S_1) + T_o \int_1^2 \frac{đQ}{T} = \int_1^2 đQ - W_{1-2} - T_o S_{\text{gen}}$$

or $$E_2 - E_1 - T_o(S_2 - S_1) = \int_1^2 đQ - W_{1-2} - T_o \int_1^2 \frac{đQ}{T} - T_o S_{\text{gen}}$$

or $$E_2 - E_1 - T_o(S_2 - S_1) = \int_1^2 \left(1 - \frac{T_o}{T}\right) \delta Q - W_{1-2} - T_o S_{\text{gen}}$$

Since for a closed system, the change in exergy between the states 1 and 2 is given by

$$\phi_2 - \phi_1 = A_2 - A_1 = E_2 - E_1 + p_o(V_2 - V_1) - T_o(S_2 - S_1)$$

$$= p_o(V_2 - V_1) + \int_1^2 \left(1 - \frac{T_o}{T}\right) \delta Q - W_{1-2} - T_o S_{\text{gen}}$$

$$A_2 - A_1 = \int_1^2 \left(1 - \frac{T_o}{T}\right) \delta Q - [W_{1-2} - p_o(V_2 - V_1)] - T_o S_{\text{gen}} \qquad (8.54)$$

Change in exergy	=	Exergy transfer with heat	–	Exergy transfer with work	+	Exergy destruction

where A or ϕ is the availability or exergy.

8.14.2 Exergy Principle

For an isolated sysem, $Q = 0$ and $W = 0$, and there is no exergy transfer with heat or work. Therefore,

$$\phi_2 - \phi_1 = A_2 - A_1 = -T_o S_{\text{gen}} = -I \qquad (8.55)$$

or in the rate form

$$\frac{d\phi}{d\tau} = \frac{dA}{d\tau} = -T_o \dot{S}_{\text{gen}} = -\dot{I} \qquad (8.56)$$

Since $\dot{I} > 0$, the only processes allowed by the second law are those for which the exergy of the isolated system decreases, or $A_2 < A_1$ and $dA/d\tau < 0$. In other words, *the exergy of an isolated system can never increase, but always decreases.*

It is the *counterpart of the entropy principle* which states that the entropy of an isolated system can never decrease, but always increases.

The exergy balance of a system can be used to determine the locations, types and magnitudes of losses (waste) of the potential of energy resources (fuels) and find ways and means to reduce these losses so as to make the system more energy efficient and for more effective use of fuel.

8.14.3 Exergy Balance for a Steady Flow System

The steady flow energy equation (S.F.E.E.) between the inlet and exit sections of the control volume (c.v.) (Fig. 8.19), is given by

First law: $$H_1 + \frac{mV_1^2}{2} + mgZ_1 + Q_{1-2}$$

$$= H_2 + \frac{mV_2^2}{2} + mgZ_2 + W_{1-2} \qquad (8.57)$$

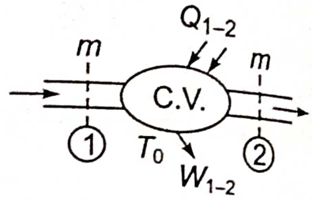

Fig. 8.19 *Exergy balance for a steady flow system*

Second law:

$$S_1 - S_2 + \int_1^2 \frac{đQ}{T} = S_{gen}$$

$S_{gen} = (S_2 - S_1 - \int \frac{dQ}{T})$

or $$T_o (S_1 - S_2) + T_o \int_1^2 \frac{đQ}{T} = T_o S_{gen} = I \qquad (8.58)$$

From Eqs. (8.57) and (8.58),

$$H_2 - H_1 - T_o(S_2 - S_1) + \frac{m(V_2^2 - V_1^2)}{2} + mg(Z_2 - Z_1)$$

$$= \int_1^2 \left(1 - \frac{T_o}{T}\right) đQ - W_{1-2} - I \qquad (8.59)$$

or $$\psi_2 - \psi_1 = A_2 - A_1 = \int_1^2 \left(1 - \frac{T_o}{T}\right) đQ - W_{1-2} - I$$

$$= m\,(a_{f2} - a_{f1}) \qquad (8.60)$$

For an isolated system,

$$\int_1^2 đQ = 0, \; W_{1-2} = 0, \; m = 0$$

$\therefore$ $$d\psi = dA = -I$$

In the rate form,

$$\sum_j \left(1 - \frac{T_o}{T}\right) \dot{Q}_j - \dot{W}_{1-2} + \dot{m}(a_{f_1} - a_{f_2}) - \dot{I}_{C.V.} = 0 \qquad (8.61)$$

where $a_{f1} - a_{f2} = h_1 - h_2 - T_o (s_1 - s_2) + \frac{V_1^2 - V_2^2}{2} + g\,(Z_1 - Z_2)$, and

$\sum_j \left(1 - \frac{T_o}{T}\right)$ is the time rate of energy transfer along with heat Q_j occurring at the location on the boundary where the instantaneous temperature is T. For a single stream entering and leaving the c.v., the exergy balance gives

$$\left(1-\frac{T_o}{T}\right)\frac{\dot{Q}}{\dot{m}}+a_{f_1}-\frac{\dot{W}}{m}-a_{f_2}=\frac{\dot{I}}{\dot{m}} \quad (8.62)$$

8.15 SECOND LAW EFFICIENCY

LO 8.12
Define second law efficiency and utilize it in various engineering devices

A common measure of energy use efficiency is the first law efficiency, η_I. The first law efficiency is defined as the ratio of the output energy of a device to the input energy of the device. The first law is concerned only with the quantities of energy, and disregards the forms in which the energy exists. It does not also discriminate between the energies available at different temperatures. It is the second law of thermodynamics which provides a means of assigning a quality index to energy. The concept of available energy or exergy provides a useful measure of energy quality (Section. 8.3).

With this concept, it is possible to analyse means of minimising the consumption of exergy to perform a given process, thereby ensuring the most efficient possible conversion of energy for the required task.

The second law efficiency, η_{II}, of a process is defined as the ratio of the minimum available energy (or exergy) which must be consumed to do a task divided by the actual amount of exergy consumed in performing the same task.

$$\eta_{II}=\frac{\text{Minimum exergy intake to perform the given task}}{\text{Actual exergy intake to perform the same task}}$$

or

$$\eta_{II}=\frac{A_{\min}}{A} \quad (8.63)$$

where A is the availability or exergy.

A power plant converts a fraction of exergy A or $W_{\max}$ to useful work W. For the desired output of W, $A_{\min}=W$ and $A=W_{\max}$. Here,

$$I=W_{\max}-W \quad \text{and} \quad \eta_{II}=\frac{W}{W_{\max}} \quad (8.64)$$

Now

$$\eta_I=\frac{W}{Q_1}=\frac{W}{W_{\max}}\cdot\frac{W_{\max}}{Q_1}$$

$$=\eta_{II}\cdot\eta_{\text{Carnot}} \quad (8.65)$$

$\therefore$

$$\eta_{II}=\frac{\eta_I}{\eta_{\text{Carnot}}} \quad (8.66)$$

Since $W_{\max}=Q_1\left(1-\frac{T_0}{T}\right)$, Eq. (8.66) can also be obtained directly as follows:

$$\eta_{II}=\frac{W}{Q_1\left(1-\frac{T_0}{T}\right)}=\frac{\eta_I}{\eta_{\text{Carnot}}}$$

If work is involved, $A_{min} = W$ (desired) and if heat is involved, $A_{min} = Q\left(1 - \frac{T_0}{T}\right)$.

If solar energy, Q_r is available at a reservoir storage temperature T_r and if quantity of heat Q_a is transferred by the solar collector at temperature T_a, then

$$\eta_I = \frac{Q_a}{Q_r}$$

and

$$\eta_{II} = \frac{\text{exergy output}}{\text{exergy input}}$$

$$= \frac{Q_a\left(1 - \frac{T_0}{T_a}\right)}{Q_r\left(1 - \frac{T_0}{T_r}\right)}$$

$$= \eta_I \frac{1 - \frac{T_0}{T_a}}{1 - \frac{T_0}{T_r}} \tag{8.67}$$

Table 8.1 shows the exergy values, and both η_I and η_{II} expressions for several common thermal tasks.

Table 8.1 $T_r > T_a > T_0 > T_c$

Task	*Energy input*	
	Input shaft work, W_i	*Q_r from reservoir at T_r*
Produce work, W_0	$A = W_i$ $A_{min} = W_0$ $\eta_I = \frac{W_0}{W_i}$ $\eta_{II} = \frac{A_{min}}{A}$ $\eta_{II} = \eta_I$ (electric motor)	$A = Q_r\left(1 - \frac{T_0}{T_r}\right)$ $A_{min} = W_0$ $\eta_I = \frac{W_0}{Q_r}$ $\eta_{II} = \eta_I \cdot \frac{1}{1 - \frac{T_0}{T_r}}$ (heat engine)

(Contd.)

Table 8.1 *(Contd.)*

Task	Energy input: Input shaft work, W_i	Energy input: Q_r from reservoir at T_r
Add heat Q_a to reservoir at T_a	$A = W_i$ $A_{min} = Q_a\left(1-\frac{T_0}{T_a}\right)$ $*\eta_I = \frac{Q_a}{W_i}$ $*\eta_{II} = \eta_I\left(1-\frac{T_0}{T_a}\right)$ (heat pump)	$A = Q_r\left(1-\frac{T_0}{T_r}\right)$ $A_{min} = Q_a\left(1-\frac{T_0}{T_a}\right)$ $\eta_I = \frac{Q_a}{Q_r}$ $\eta_{II} = \eta_I \frac{1-\frac{T_0}{T_a}}{1-\frac{T_0}{T_r}}$ (solar water heater)
Extract heat Q_c from cold reservoir at T_c (below ambient)	$A = W_i$ $A_{min} = Q_c\left(\frac{T_0}{T_c}-1\right)$ $*\eta_I = \frac{Q_c}{W_i}$ $*\eta_{II} = \eta_I\left(\frac{T_0}{T_c}-1\right)$ (Refrigerator—electric motor driven)	$A = Q_r\left(1-\frac{T_0}{T_r}\right)$ $A_{min} = Q_c\left(\frac{T_0}{T_c}-1\right)$ $*\eta_I = \frac{Q_c}{Q_r}$ $*\eta_{II} = \eta_I\left(\frac{\frac{T_0}{T_c}-1}{1-\frac{T_0}{T_r}}\right)$ (Refrigerator—heat operated)

In the case of a heat pump, the task is to add heat Q_a to a reservoir to be maintained at temperature T_a and the input shaft work is W_i.

$$\text{COP} = \frac{Q_a}{W_i} = \eta_I, \text{ say}$$

$$(\text{COP})_{max} = \frac{T_a}{T_a - T_0} = \frac{Q_a}{W_i} = \frac{Q_a}{A_{min}}$$

$$\therefore \quad A_{min} = Q_a\left(1-\frac{T_0}{T_a}\right)$$

$$\therefore \quad \eta_{II} = \frac{A_{min}}{A} = \frac{Q_a\left(1-\dfrac{T_0}{T_a}\right)}{W_i}$$

$$\eta_{II} = \eta_I\left(1-\frac{T_0}{T_a}\right) \tag{8.68}$$

Similarly, expressions of η_I and η_{II} can be obtained for other thermal tasks.

8.15.1 Matching end use to Source

Combustion of a fuel releases the necessary energy for the tasks such as space heating, process steam generation and heating in industrial furnaces. When the products of combustion are at a temperature much greater than that required by a given task, the end use is not well matched to the source and results in inefficient utilisation of the fuel burned. To illustrate this, let us consider a closed system receiving a heat transfer $\dot{Q}_r$ at a source temperature T_r and delivering $\dot{Q}_a$ at a use temperature T_a (Fig. 8.20). Energy is lost to the surroundings by heat transfer at a rate $\dot{Q}_l$ across a portion of the surface at T_l. At a steady state, the energy and exergy rate balances become

$$\dot{Q}_r = \dot{Q}_a + \dot{Q}_l \tag{8.69}$$

$$Q_r\left(1-\frac{T_0}{T_r}\right) = Q_a\left(1-\frac{T_0}{T_a}\right) + \dot{Q}_l\left(1-\frac{T_0}{T_l}\right) + \dot{I} \tag{8.70}$$

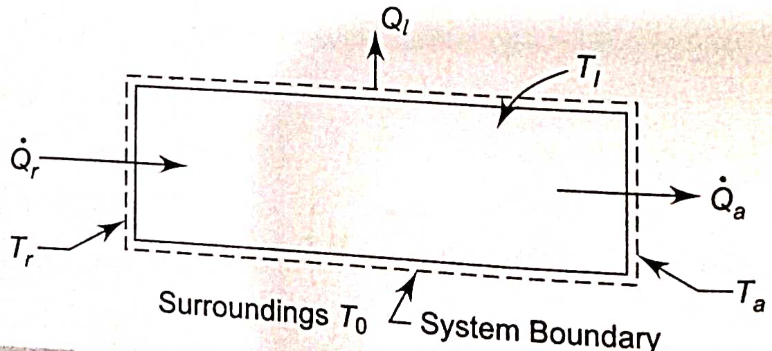

Fig. 8.20 *Efficient energy utilisation from second law viewpoint*

Equation (8.69) indicates that the energy carried in by heat transfer $\dot{Q}_r$ is either used, $\dot{Q}_a$, or lost to the surroundings, $\dot{Q}_l$. Then

$$\eta_I = \frac{\dot{Q}_a}{\dot{Q}_r} \tag{8.71}$$

The value of η_I can be increased by increasing insulation to reduce the loss. The limiting value, when $\dot{Q}_l = 0$, is $\eta_I = 1$ (100 percent).

Equation (8.70) shows that the exergy carried into the system accompanying the heat transfer $\dot{Q}_r$ is either transferred from the system accompanying the heat transfer $\dot{Q}_a$ and $\dot{Q}_l$ or destroyed by irreversibilities within the system, $\dot{I}$.

Therefore,

$$\eta_{II} = \frac{\dot{Q}_a\left(1-\frac{T_0}{T_a}\right)}{\dot{Q}_r\left(1-\frac{T_0}{T_r}\right)} = \eta_I \frac{1-\frac{T_0}{T_a}}{1-\frac{T_0}{T_r}} \tag{8.72}$$

Both η_I and η_{II} indicate how effectively the input is converted into the product. The parameter η_I does this on energy basis, whereas η_{II} does it on an exergy basis.

For proper utilisation of exergy, it is desirable to make η_I as close to unity as practical and also a good match between the source and use temperatures, T_r and T_a. Figure 8.21 demonstrates the second law efficiency against the use temperature T_a for an assumed source temperature T_r = 2200 K. It shows that η_{II} tends to unity (100 percent) as T_a approaches T_r. The lower the T_a, the lower becomes the value of η_{II}. Efficiencies for three applications, viz., space heating at T_a = 320 K, process steam generation at T_a = 480 K, and heating in industrial furnaces at T_a = 700 K, are indicated in the figure. It suggests that fuel is used far more effectively in the high temperature use. An excessive temperature gap between T_r and T_a causes a low η_{II} and an inefficient energy utilisation. A fuel or any energy source is consumed efficiently when the first user temperature approaches the fuel temperature. This means that the fuel should first be used for high temperature applications. The heat rejected from these applications can then be cascaded to applications at lower temperatures, eventually to the task of, say, keeping a building warm. This is called *energy cascading* and ensures more efficient energy utilisation.

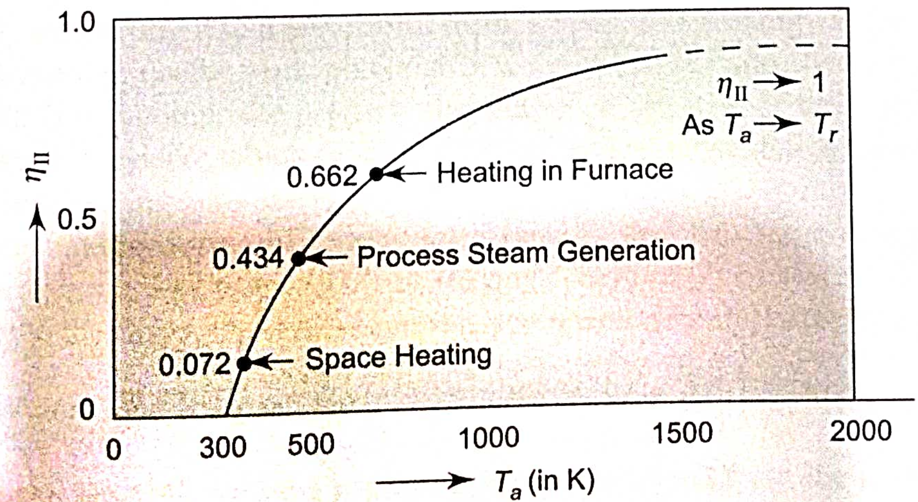

Fig. 8.21 *Effect of use temperature T_a on the second law efficiency (T_r = 2200 K, T_0 = 300 K, η_I = 100 percent)*

8.15.2 Further Illustrations of Second Law Efficiencies

Second law efficiency of different components can be expressed in different forms. It is derived by using the exergy balance rate given as follows:

1. Turbines The steady state exergy balance (Fig. 8.22) gives

$$\frac{\dot{Q}}{\dot{m}}\left[1-\frac{T_0}{T_\sigma}\right]+a_{f_1}=\frac{\dot{W}}{\dot{m}}+a_{f_2}+\frac{\dot{I}}{\dot{m}}$$

where $a_{f1} = h_1 - T_o s_1$ and $a_{f2} = h_2 - T_o s_2$

If there is no heat loss,

$$a_{f1}-a_{f2}=\frac{\dot{W}}{\dot{m}}+\frac{\dot{I}}{\dot{m}} \quad (8.73)$$

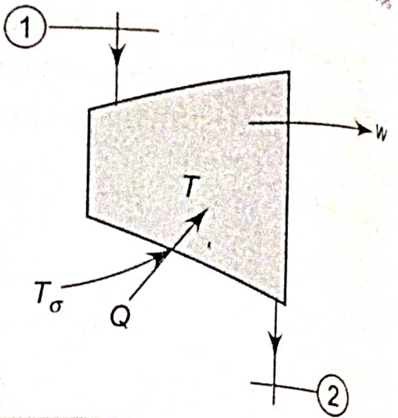

Fig. 8.22 *Exergy balance of a turbine*

The second law efficiency,

$$\eta_{II}=\frac{\dot{W}/\dot{m}}{a_{f_1}-a_{f_2}} \quad (8.74)$$

2. Compressor and pump Similarly, for a compressor or a pump,

$$\frac{\dot{W}}{\dot{m}}=a_{f2}-a_{f1}+\frac{\dot{I}}{\dot{m}}$$

and

$$\eta_{II}=\frac{a_{f_2}-a_{f_2}}{-\dot{W}/\dot{m}} \quad (8.75)$$

3. Heat exchanger Writing the exergy balance for the heat exchanger, (Fig. 8.23)

$$\sum\left[1-\frac{T_0}{T_j}\right]\dot{Q}_j-\dot{W}_{\text{C.V.}}+[\dot{m}_h\, a_{f_1}+\dot{m}_c\, a_{f_3}]-[\dot{m}_h\, a_{f_2}+\dot{m}_c\, a_{f_4}]-\dot{I}_{\text{C.V.}}=0\ \text{in}_c$$

If there is no heat transfer and work transfer,

$$\dot{m}_h\,[a_{f1}-a_{f2}]=\dot{m}_c\,[a_{f4}-a_{f3}]+\dot{I} \quad (8.76)$$

$$\eta_{II}=\frac{\dot{m}_c\,[a_{f_4}-a_{f_3}]}{\dot{m}_h\,[a_{f_1}-a_{f_2}]} \quad (8.77)$$

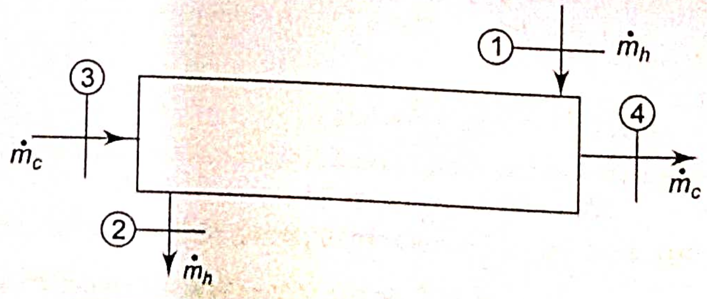

Fig. 8.23 *Exergy balance of a heat exchanger*

4. Mixing of two fluids Exergy balance for the mixer (Fig. 8.24) gives

$$\left[1-\frac{T_0}{T_\sigma}\right]\dot{Q}+\dot{m}_1 a_{f_1}+\dot{m}_2 a_{f_2}=\dot{W}_{\text{C.V.}}+\dot{m}_3\, a_{f_3}+I_{\text{C.V.}}$$

If the mixing is adiabatic and since $\dot{W}_{C.V.} = 0$ and $\dot{m}_1 + \dot{m}_2 = \dot{m}_3$.

$$\dot{m}_1[a_{f1} - a_{f3}] = \dot{m}_2[a_{f3} - a_{f2}] + \dot{I} \quad (8.78)$$

and

$$\eta_{II} = \frac{\dot{m}_2[a_{f_3} - a_{f_2}]}{\dot{m}_1[a_{f_1} - a_{f_3}]} \quad (8.79)$$

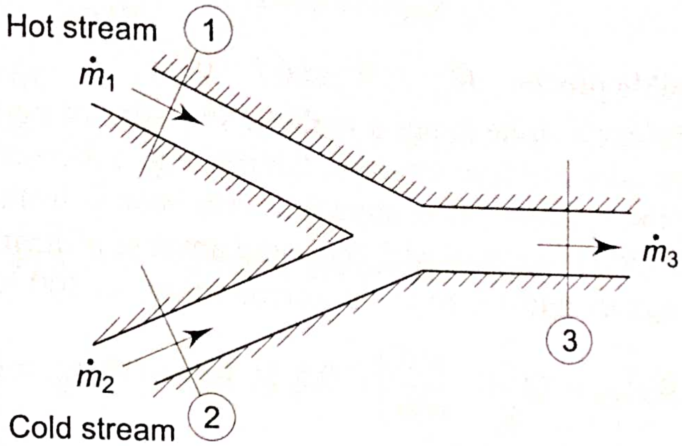

Fig. 8.24 *Exergy loss due to mixing*

8.16 COMMENTS ON EXERGY

The energy of the universe, like its mass, is constant. Yet at times, we are bombarded with speeches and articles on how to 'conserve' energy. As engineers, we know that energy is always conserved. What is not conserved is the exergy, i.e. the useful work potential of the energy. Once the exergy is wasted, it can never be recovered. When we use energy (electricity) to heat our homes, we are not destroying any energy, we are merely converting it to a less useful form, a form of less exergy value.

The maximum useful work potential of a system at the specified state is called exergy which is a composite property depending on the state of the system and the surroundings. A system which is in equilibrium with its surroundings is said to be at the dead state having zero exergy.

The mechanical forms of energy such as K.E. and P.E. are entirely available energy or exergy. The exergy (W) of thermal energy (Q) of reservoirs (TER) is equivalent to the work output of a Carnot heat engine operating between the reservoir at temperature T and environment at T_0, i.e. $W = Q\left[1 - \frac{T_0}{T}\right]$.

The actual work W during a process can be determined from the first law. If the volume of the system changes during a process, part of this work (W_{surr}) is used to push the surrounding medium at constant pressure p_0 and it cannot be used for any useful purpose. The difference between the actual work and the surrounding work is called *useful work*, W_u

$$W_u = W - W_{surr} = W - p_0(V_2 - V_1)$$

W_{surr} is zero for cyclic devices, for steady flow devices and for system with fixed boundaries (rigid walls).

The maximum amount of useful work that can be obtained from a system as it undergoes a process between two specified states is called *reversible work*, W_{rev}. If the final state of the system is the dead state, the reversible work and the exergy become identical.

The difference between the reversible work and useful work for a process is called *irreversibility*.

$$I = W_{rev} - W_u = T_0\, S_{gen}$$

$$\dot{I} = T_0\, \dot{S}_{gen}$$

For a total reversible process, $W_{rev} = W_u$ and $I = 0$.

The first law efficiency alone is not a realistic measure of performance for engineering devices. Consider two heat engines, having, e.g. a thermal efficiency of, say, 30 percent. One of the engines (A) is supplied with heat Q from a source at 600 K and the other engine (B) is supplied with the same amount of heat Q from a source at 1000 K. Both the engines reject heat to the surroundings at 300 K.

$$(W_A)_{rev} = Q\left(1 - \frac{300}{600}\right) = 0.5\,Q, \text{ while } (W_A)_{act} = 0.3Q$$

Similarly,

$$(W_B)_{rev} = Q\left(1 - \frac{300}{1000}\right) = 0.7\,Q, \text{ and } (W_B)_{act} = 0.3Q$$

At first glance, both engines seem to convert the same fraction of heat, that they receive, to work, thus performing equally well from the viewpoint of the first law. However, in the light of second law, the engine B has a greater work potential (0.7 Q) available to it and thus should do a lot better than engine A. Therefore, it can be said that engine B is performing poorly relative to engine A, even though both have the same thermal efficiency.

To overcome the deficiency of the first law efficiency, a second law efficiency hII can be defined as the ratio of actual thermal efficiency to the maximum possible thermal efficiency under the same conditions:

$$\eta_{II} = \frac{\eta_I}{\eta_{rev}}$$

So, for engine A, $\eta_{II} = 0.3/0.5 = 0.60$

and for engine B, $\eta_{II} = 0.3/0.7 = 0.43$

Therefore, the engine A is converting 60 percent of the available work potential (exergy) to useful work. This is only 43 percent for the engine B. Therefore,

$$\eta_{II} = \frac{\eta_{act}}{\eta_{rev}} = \frac{W_u}{W_{rev}} \text{ (for heat engines and other work producing devices)}$$

$$\eta_{II} = \frac{COP}{COP_{rev}} = \frac{W_{rev}}{W_u} \text{ (for refrigerators, heat pumps and other work absorbing}$$

devices).

The exergies of a closed system (ϕ) and a flowing fluid system (ψ) are given on unit mass basis:

$$\phi = (u - u_0) - T_0(s - s_0) + p_0(v - v_0) \text{ kJ/kg}$$

$$\psi = (h - h_0) - T_0(s - s_0) + \frac{V^2}{2} + gz \text{ kJ/kg}$$

8.16.1 Reversible Work Expressions

1. Cyclic devices

$$W_{rev} = \eta_{rev}\, Q_1 \text{ (Heat engines)}$$

$$-W_{rev} = \frac{Q_2}{(COP_{rev})_{Ref.}} \text{ (Refrigerators)}$$

$$-W_{rev} = \frac{Q_2}{(COP_{rev})_{HP}} \text{ (Heat pumps)}$$

2. Closed system

$$W_{rev} = U_1 - U_2 - T_0(S_1 - S_2) + p_0(V_1 - V_2)$$

$$= m(\phi_1 - \phi_2)$$

3. Steady flow system (single stream)

$$W_{rev} = m\left[\left(h_1 + \frac{V_1^2}{2} + gz_1 - T_0 s_1\right) - \left(h_2 + \frac{V_2^2}{2} + gz_2 - T_0 s_2\right)\right]$$

$$= m(\psi_1 - \psi_2)$$

When the system exchanges heat with another reservoir at temperature T_k other than the atmosphere,

$$\dot{W}_{rev} = \dot{m}(\psi_1 - \psi_2) + \dot{Q}_k\left(1 - \frac{T_0}{T_k}\right)$$

The first law efficiency is defined as the ratio of energy output and energy input, while their difference is the energy loss. Likewise, the second law efficiency is defined as the ratio of exergy output and exergy input and their difference is irreversibility. By reducing energy loss, first law efficiency can be improved. Similarly, by reducing irreversibilities, the second law efficiency can be enhanced.